MERCURY: Metaphor Evidence Reasoning with Cross-View Consistency and Equivariance-Constrained Calibration

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Abstract

Multimodal metaphor understanding is a demanding test of deep semantic reasoning in multimodal large language models (MLLMs), requiring the integration of visual cues, embedded text, emotional tone, and cultural context to derive abstract, non-literal meanings. Existing resources, however, focus primarily on metaphor recognition, attribute annotation, or open-ended interpretation generation, making it difficult to evaluate whether models can discriminate among closely related interpretations. To address this gap, we introduce BiMeRe, a Chinese-English bilingual benchmark for fine-grained multimodal metaphor understanding comprising 6,068 curated metaphorical samples across diverse visual genres. Each sample is formulated as a six-choice interpretation selection task with human-verified answers and rationale annotations. Evaluation of 18 representative open-source and closed-source MLLMs reveals that strong vision-language capabilities do not guarantee reliable metaphor interpretation, and that Chain-of-Thought (CoT), domain-oriented, and emotion-oriented prompting strategies do not consistently improve performance. To improve reasoning stability, we propose MERCURY, which performs two-stage task adaptation via visual-semantic alignment and metaphor-reasoning instruction tuning, employs Metaphor-Specialized Multi-View Reasoning to acquire complementary evidence from anchor, textual-interaction, and challenge views, and applies Equivariance-Constrained Cross-View Evidence Calibration to revise predictions only when cross-view consistency, semantic preservation, and option-order stability are jointly satisfied. MERCURY achieves accuracies of 70.87% and 72.07% on the English and Chinese test sets, respectively, outperforming all evaluated MLLMs. The dataset and code are not provided at this stage to preserve author anonymity and will be made publicly available upon publication. Code is available at: https://github.com/constant-lou/MERCURY,

CCS Concepts

• **Computing methodologies** → **Computer vision tasks**; **Language resources**.

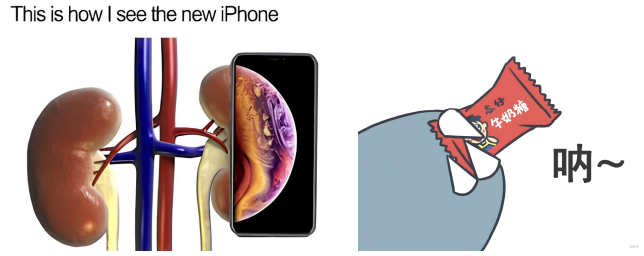
Keywords

Multimodal metaphor understanding, Multimodal large language model, Bilingual multimodal benchmark

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(a) A metaphor linking iPhone and kidney (b) A Chinese metaphor about homophony

Figure 1: Sample metaphors from the BiMeRe.

1 Introduction

Metaphor is a fundamental cognitive mechanism that conveys complex emotional and cultural meanings through mappings between source and target domains [11, 13]. In digital media, metaphors are increasingly constructed through interacting visual and textual cues [4, 9, 19]. As shown in Figure 1(a), the juxtaposition of an iPhone and a kidney conveys high cost and personal sacrifice. Figure 1(b) illustrates a Chinese homophonic metaphor in which the brand name *Wangzai* is reinterpreted as *wang zai*, literally "forgetting the child," evoking forgetting and emotional loss.

Despite their strong performance on general vision-language tasks, multimodal large language models (MLLMs) still struggle to connect concrete visual cues with abstract and culturally grounded metaphorical meanings. Existing datasets mainly emphasize metaphor recognition, attribute classification, or open-ended interpretation, and provide limited support for evaluating fine-grained discrimination among plausible meanings. To address this gap, we introduce BiMeRe, a Chinese–English bilingual benchmark containing 6,068 curated metaphorical samples, each formulated as a six-choice interpretation task. Our evaluation of 18 representative MLLMs further shows that standard Chain-of-Thought and task-oriented prompting do not consistently improve performance, as unconstrained reasoning can amplify literal interpretation bias and introduce associations that are insufficiently grounded in visual evidence.

To address these challenges, we propose MERCURY, a framework for multimodal metaphor understanding that combines structured evidence reasoning with conservative decision calibration. MERCURY first performs two-stage task adaptation to strengthen

metaphor-oriented visual-semantic alignment and reasoning ability. It then acquires complementary evidence from anchor, textual-interaction, and challenge views, allowing the model to examine global metaphorical meaning, image-text relations, and potential distractors from distinct perspectives. Finally, cross-view consistency, semantic preservation, and option-permutation equivariance are jointly used to constrain answer revision. This design improves reasoning stability while reducing unreliable prediction changes caused by literal bias, semantic drift, and option-order sensitivity. The main contributions of this study include the following:

- We introduce BiMeRe, a Chinese-English bilingual benchmark containing 6,068 curated multimodal metaphor samples, formulated as a six-choice interpretation task with human-verified answers and reasoning annotations.
- We propose MERCURY, which combines multi-view evidence reasoning with equivariance-constrained calibration to improve metaphor interpretation through cross-view consistency, semantic preservation, and option-permutation stability.
- We evaluate 18 representative MLLMs on BiMeRe. MERCURY achieves 70.87% and 72.07% accuracy on the English and Chinese test sets, respectively, while ablation and mechanism analyses confirm the effectiveness of its core components.

2 Related Work

2.1 Metaphor Understanding Datasets

Research on metaphor understanding has expanded from textual metaphor recognition to multimodal settings involving images, embedded text, and cultural context. Existing datasets, including MultiMET, MET-Meme, NYK-MS, V-FLUTE, DEEPEVAL, II-Bench, and CII-Bench, support tasks such as metaphor detection, attribute classification, visual rhetoric analysis, and cross-cultural interpretation [5, 17, 20, 24, 26, 27, 30, 31]. However, they provide limited evaluation of fine-grained discrimination among semantically similar interpretations. BiMeRe addresses this limitation through a Chinese-English six-choice task with human-verified answers and reasoning annotations.

2.2 Multimodal Metaphor Reasoning

Chain-of-Thought prompting and its multimodal extensions, such as Multimodal-CoT, DDCoT, and C4MMD, improve reasoning by connecting visual observations with intermediate semantic inference [22, 25, 32, 33]. Nevertheless, single-path reasoning can amplify literal bias, unsupported associations, and sensitivity to option wording or order. MERCURY instead gathers complementary evidence from anchor, textual-interaction, and challenge views, and performs conservative prediction revision through cross-view consistency, semantic preservation, and option-permutation verification.

3 The BiMeRe Dataset

3.1 Data Construction and Annotation

BiMeRe is constructed by aggregating metaphorical images from several established datasets. The Chinese subset is collected from MET-Meme (78%) and CII-Bench (22%), while the English subset is collected from MET-Meme (66%), II-Bench (18%), and DEEPEVAL (16%) [17, 24, 27, 30]. We retain only samples containing clear metaphorical expressions and apply manual deduplication and quality screening, resulting in 6,068 curated metaphorical samples.

Each sample contains a metaphorical image, a standardized question, six candidate interpretations, and a detailed reasoning annotation. Among the six candidates, one is the correct interpretation and the remaining five are semantically plausible distractors. For samples inherited from II-Bench and CII-Bench, we preserve the original core annotations while manually revising ambiguous options and correcting labels when necessary. This design transforms metaphor understanding into a fine-grained interpretation-selection task rather than a simple detection or classification problem.

3.2 Quality Control and Reasoning Annotation

We employ a human-AI collaborative annotation pipeline. An LLM first generates candidate interpretations and explanatory rationales, which are subsequently reviewed and revised by human annotators. To reduce annotation artifacts, we monitor template repetition, intra-sample option balance, inter-sample semantic diversity, and distractor quality. Samples that violate the predefined quality criteria undergo targeted manual revision. Detailed metrics, thresholds, and annotation prompts are provided in the supplementary material.

Reasoning annotations are generated using GPT-4o and manually verified for logical consistency, interpretation accuracy, and linguistic quality. The rationales cover the main reasoning process from visual cue recognition and image-text interaction to candidate comparison and final metaphor interpretation. The resulting average annotation quality score is $Q_{\text{total}} = 0.79$, satisfying the predefined quality requirement.

3.3 Dataset Statistics and Human Evaluation

BiMeRe covers diverse image types, including memes, illustrations, comics, paintings, posters, and logos. It also spans multiple source domains, emotional orientations, and difficulty levels. As shown in Figure 2, the Chinese and English subsets exhibit different distributions across these dimensions, reflecting language- and culture-specific patterns in multimodal metaphorical expression.

To assess annotation reliability and establish a human benchmark, we sample 200 examples from each test subset using stratified random sampling. Annotators complete the task independently under blind conditions, and final judgments are determined by majority voting. Cases of complete three-way disagreement, accounting for 2% of the samples, are adjudicated by an additional annotator. Fleiss' kappa reaches 0.81 for the English subset and 0.78 for the Chinese subset, indicating substantial inter-annotator agreement [14]. Human accuracy reaches 92.45% and 87.21% on the English and Chinese subsets, respectively.

4 MERCURY

Existing MLLMs are primarily pretrained on general image-text pairs, with visual-semantic alignment optimized for object recognition, image captioning, and literal visual question answering.

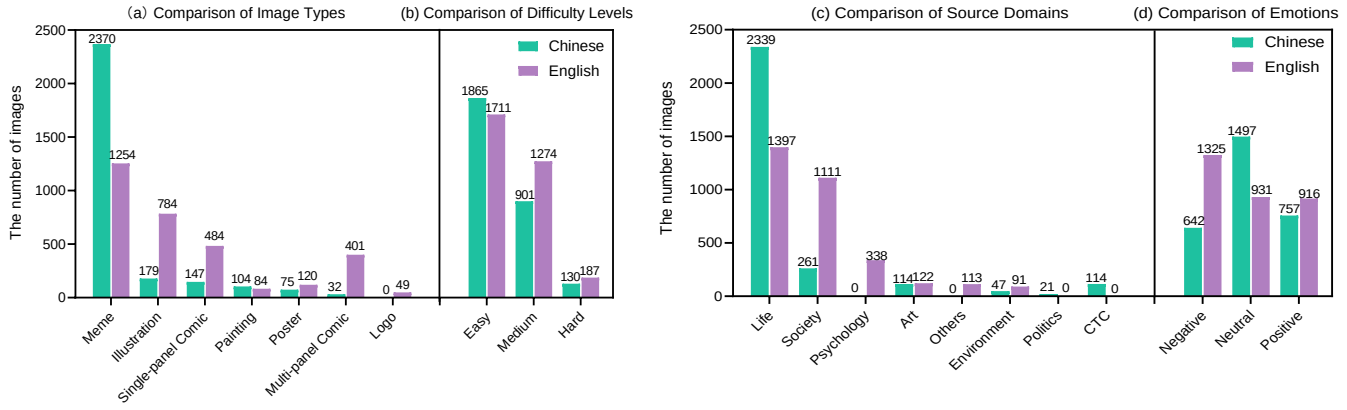


Figure 2: Statistical distributions of the Chinese and English subsets of BiMeRe across image type, difficulty, source domain, and emotional orientation.

However, multimodal metaphor understanding requires models to connect concrete visual cues with abstract meanings by integrating visual entities, embedded text, scene relations, emotional tone, and cultural context. Consequently, single-path reasoning may overemphasize literal or local evidence and introduce interpretations that are insufficiently grounded in the image.

To address these challenges, we propose MERCURY, a structured evidence reasoning and conservative calibration framework for Chinese-English multimodal metaphor understanding. As shown in Figure 3, MERCURY first adopts a two-stage task adaptation strategy to strengthen visual-semantic alignment and metaphor reasoning. It then employs Metaphor-Specialized Multi-View Reasoning (MS-MVR) to gather complementary evidence from the Anchor, Textual Interaction, and Challenge views. Finally, Equivariance-Constrained Cross-View Evidence Calibration (EC-CVEC) revises predictions conservatively according to cross-view consistency, semantic preservation, and option-permutation stability.

MERCURY follows a standard vision-language architecture consisting of a visual encoder, a cross-modal projector, and a large language model. We use GIT-Large-TextVQA [21] as the visual encoder due to its OCR and fine-grained visual-semantic representation capabilities. The resulting 1,024-dimensional visual features are mapped into the 5,120-dimensional language embedding space through a two-layer MLP projector with GELU activation. The projected visual tokens are combined with the textual input and processed by Vicuna-13B-v1.5 [34] to generate reasoning evidence and the final answer.

4.1 Task Definition

We formulate multimodal metaphor understanding as a multiple-choice reasoning task. Given an image I, a question Q, and a candidate interpretation set $\mathcal{Y}$, the model generates a reasoning chain R and predicts the answer $A \in \mathcal{Y}$:

$$(\hat{R}, \hat{A}) = \underset{R,\, A \in \mathcal{Y}}{\operatorname{argmax}} P(R, A \mid I, Q, \mathcal{Y}; \Theta), \tag{1}$$

where Θ denotes the model parameters. Each BiMeRe sample contains six candidate interpretations. Unlike conventional visual question answering, the task requires non-literal semantic inference and fine-grained discrimination among semantically related interpretations.

4.2 Two-Stage Task Adaptation

Multimodal metaphor understanding requires both reliable visual-semantic alignment and task-specific non-literal reasoning. MERCURY therefore adopts a progressive two-stage adaptation strategy that separates cross-modal representation learning from metaphor reasoning instruction tuning.

Stage 1: Visual-Semantic Alignment. The first stage adapts the cross-modal projector to metaphorical and text-rich images. The visual encoder and language model are frozen, while only the projector is optimized. Given the projected visual representation H_I and a target visual-description sequence $X = (x_1, \ldots, x_T)$, the training objective is

$$\mathcal{L}_1 = -\sum_{t=1}^{T} \log P(x_t \mid x_{<t}, H_I). \tag{2}$$

This stage strengthens the alignment of visual entities, embedded text, local relations, and scene-level information with the language model's semantic space, providing a reliable perceptual foundation for subsequent metaphor reasoning.

Stage 2: Metaphor Reasoning Instruction Tuning. In the second stage, the visual encoder remains frozen, while the projector and Vicuna are jointly optimized on BiMeRe. Let R_i and A_i denote the ground-truth reasoning sequence and answer identifier of the i-th training sample, respectively. Let Θ_2 denote the trainable parameters of the projector and Vicuna in this stage. The instruction-tuning objective is

$$\mathcal{L}_2 = -\sum_{i=1}^{N} \log P(R_i, A_i \mid I_i, Q_i, \mathcal{Y}_i; \Theta_2). \tag{3}$$

This stage enables the model to identify metaphor-relevant visual cues, integrate image-text relations, compare candidate interpretations, and generate evidence consistent with its prediction. After the two stages, we obtain the task-adapted model M_{FT}, which serves as the backbone for MS-MVR and EC-CVEC.

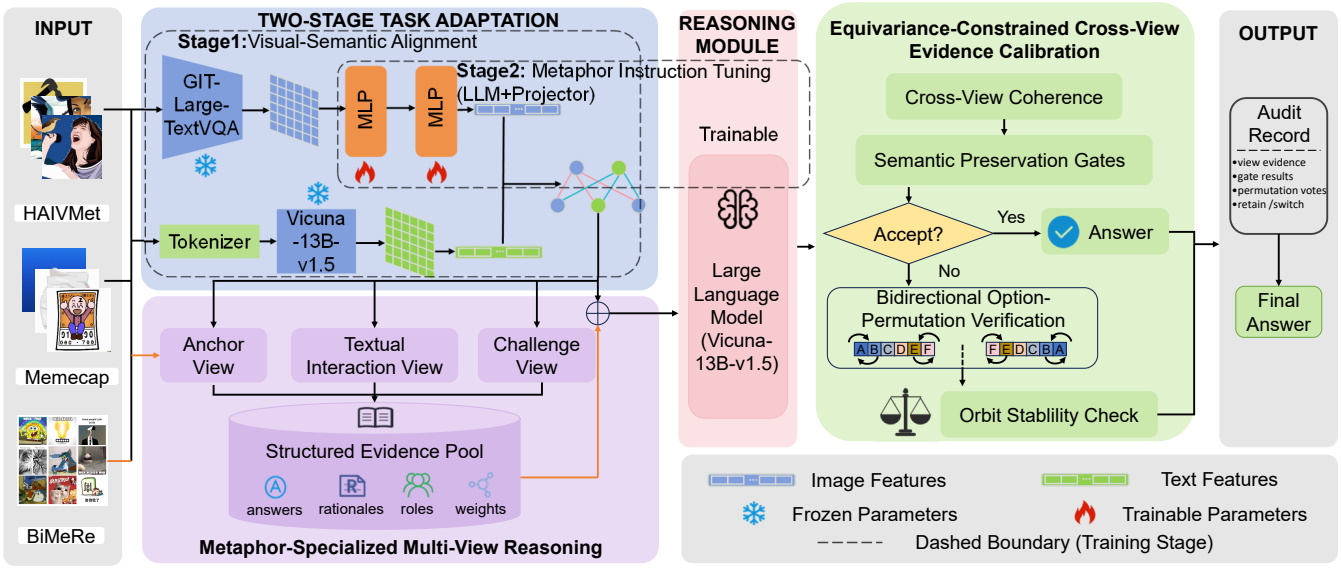


Figure 3: Overview of the proposed MERCURY framework.

4.3 Metaphor-Specialized Multi-View Reasoning

Building on the task-adapted model M_{FT}, MERCURY further introduces Metaphor-Specialized Multi-View Reasoning (MS-MVR) to acquire complementary evidence before making the final decision. Although M_{FT} has learned basic metaphor reasoning, a single inference path may still overemphasize salient visual objects, local text, or the wording of candidate interpretations. Multimodal metaphor understanding instead requires the joint consideration of global visual meaning, image-text interaction, and potentially misleading alternatives.

Unlike conventional self-consistency methods based on random sampling, MS-MVR employs three deterministic views with distinct functional roles:

- **Anchor View.** This view identifies the main visual entities, their relations, and the most direct source-to-target mapping. It provides a global interpretation grounded in the core semantics of the image.
- **Textual Interaction View.** This view examines how embedded text interacts with visual content, including OCR text, labels, slogans, irony, and homophonic wordplay that may alter or enrich the metaphorical meaning.
- **Challenge View.** This view checks whether the current interpretation relies on literal meaning, superficial association, or semantically plausible but incorrect distractors.

Let $\mathcal{V}_\ell$ denote the set of reasoning views for language ℓ. For each view $v \in \mathcal{V}_\ell$, M_{FT} generates a reasoning chain $\hat{R}_v$ and predicts an answer $\hat{A}_v \in \mathcal{Y}$ under the corresponding view prompt p_v:

$$(\hat{R}_v, \hat{A}_v) = M_{\text{FT}}(I, Q, \mathcal{Y}; p_v). \tag{4}$$

Each view is assigned a functional role $\rho(v) \in \{\text{A}, \text{T}, \text{C}\}$, representing the Anchor, Textual Interaction, and Challenge roles, respectively, together with a non-negative reliability weight w_v. These weights are selected on the validation set and remain fixed during testing. For each candidate interpretation $a \in \mathcal{Y}$, MS-MVR computes its weighted support and selects the initial multi-view answer as

$$W(a) = \sum_{v \in \mathcal{V}_\ell} w_v \mathbb{I}[\hat{A}_v = a], \qquad A_b = \operatorname*{argmax}_{a \in \mathcal{Y}} W(a). \tag{5}$$

where $\mathbb{I}[\cdot]$ is the indicator function and A_b denotes the multi-view baseline answer. Ties are resolved using a deterministic rule fixed on the validation set.

In addition to A_b, MS-MVR retains the generated reasoning chain, predicted answer, functional role, and reliability weight of each view. For a candidate a, its supporting role set is denoted by $\mathcal{R}(a) = \{\rho(v) \mid v \in \mathcal{V}_\ell, \hat{A}_v = a\}$. These outputs form a structured evidence pool that records both agreement and disagreement across views, providing the basis for the subsequent equivariance-constrained calibration stage.

4.4 Equivariance-Constrained Cross-View Evidence Calibration

Although MS-MVR collects complementary evidence from multiple views, weighted aggregation may still be affected by repeated surface cues, weakly grounded interpretations, and option-position bias. We therefore introduce Equivariance-Constrained Cross-View Evidence Calibration (EC-CVEC), which retains the multi-view baseline answer A_b by default and revises it only when an alternative candidate passes additional evidence and stability checks.

Cross-View Calibration. Based on the weighted support $W(a)$ and supporting role set $\mathcal{R}(a)$ produced by MS-MVR, EC-CVEC considers only alternatives that have been predicted by at least one reasoning view. An alternative is regarded as cross-view consistent when either: (1) it is supported by the Anchor View and its weighted support is sufficiently close to that of A_b; or (2) it is jointly supported by the Textual Interaction and Challenge views and ties the support of A_b.

Table 1: Main evaluation results on the BiMeRe dataset under None (Zero-shot) settings.

	en										zh									
Models	**Overall**	**Life**	**Art**	**Society**	**Env**	**Psy**	**Others**	**Positive**	**Negative**	**Neutral**	**Overall**	**Life**	**Art**	**Society**	**Env**	**Politics**	**CTC**	**Positive**	**Negative**	**Neutral**
Open-source Models																				
VisualGLM-6B	16.87	16.96	26.23	16.11	23.08	14.79	14.29	14.63	15.77	20.62	19.72	20.09	22.81	14.56	21.28	14.29	21.05	18.89	16.67	21.44
InternLM-XComposer2-VL-7B	33.42	34.07	42.62	32.85	27.47	29.59	36.61	36.46	28.91	36.84	27.24	28.90	21.93	19.54	25.53	14.29	19.30	28.53	22.12	28.79
Qwen-VL-Chat	36.41	37.72	37.70	34.02	32.97	33.73	52.68	37.34	35.25	37.16	43.13	45.79	27.19	32.57	44.68	42.86	28.07	42.80	41.59	43.95
InstructBLIP-T5-XL	43.32	43.81	48.36	41.31	49.45	42.90	47.32	44.21	40.30	46.72	-	-	-	-	-	-	-	-	-	-
BLIP-2-Flan-T5-XL	45.87	45.74	56.56	44.64	49.45	42.60	54.46	46.94	43.09	48.76	-	-	-	-	-	-	-	-	-	-
Yi-VL-6B-Chat	46.53	46.89	63.11	44.82	45.05	43.79	50.00	45.74	44.00	50.91	50.41	52.89	42.98	37.55	51.06	47.62	36.84	52.71	43.77	52.10
MiniCPM-Llama3-V-2.5	49.27	52.83	59.02	44.01	52.75	46.75	50.89	51.20	45.13	53.28	44.16	49.08	19.30	28.74	25.53	33.33	13.16	46.37	36.76	46.23
Idefics2-8B	51.58	51.75	54.92	50.41	54.95	49.70	59.82	50.22	50.26	54.78	38.16	38.09	41.23	34.87	51.06	57.14	35.09	40.37	33.24	41.49
Yi-VL-34B-Chat	54.63	55.55	63.11	51.49	59.34	53.55	64.29	57.21	50.04	58.65	59.25	62.08	49.12	45.21	57.45	57.14	44.74	60.24	55.61	60.32
DeepSeek-VL2	55.33	55.76	72.13	52.66	48.35	55.03	64.29	53.28	51.62	62.62	62.50	65.93	41.23	49.81	63.83	52.38	43.86	61.29	60.28	64.06
InternVL2-8B	56.37	56.19	68.03	54.64	58.24	55.33	64.29	56.11	53.66	60.47	65.19	68.28	45.61	53.64	70.21	52.38	48.25	63.28	63.08	67.07
InternVL-2.5-8B	58.51	57.77	69.67	57.79	59.34	58.58	61.61	56.11	57.06	62.94	66.64	69.56	50.88	57.85	61.70	57.14	46.49	65.65	66.51	67.20
Qwen2.5-VL-72B	63.32	64.14	74.17	60.53	60.44	65.27	65.14	61.84	61.24	67.71	66.57	69.06	55.26	**60.62**	60.87	57.14	44.74	67.60	65.52	66.51
Closed-source Models																				
GPT-4o	62.04	61.42	68.85	60.67	58.24	64.79	70.54	62.26	60.53	65.95	59.60	62.25	43.86	52.11	55.32	61.90	39.47	57.73	60.28	60.25
Qwen-VL-Max	62.73	63.41	72.50	60.49	57.14	62.87	69.72	61.79	60.35	64.61	66.92	69.58	56.14	57.53	58.70	**76.19**	46.49	68.92	65.99	66.31
Gemini-2.5-Flash	63.75	62.71	68.85	63.46	**71.43**	63.61	67.86	59.06	61.66	71.32	51.55	52.46	52.63	46.36	48.94	61.90	42.98	53.90	51.71	50.30
GLM-4V-Plus	64.53	63.78	66.39	63.46	**71.43**	66.57	70.54	65.17	62.42	66.92	65.09	66.95	57.02	54.41	**72.34**	61.90	**57.02**	67.90	62.93	64.60
Claude-3.7-Sonnet	66.20	66.55	**78.51**	63.94	62.64	**66.87**	**71.43**	67.95	61.13	**71.66**	65.24	67.55	50.88	60.15	51.11	71.43	48.25	65.87	64.64	65.17
MERCURY(Ours)	**70.87**	**73.07**	75.00	**66.65**	-	-	-	**73.97**	**69.59**	65.69	**72.07**	**77.16**	**71.00**	53.33	60.56	59.14	45.00	**72.95**	**67.92**	**71.09**
Human Performance																				
Human	92.45	91.62	90.94	94.55	95.92	89.96	93.88	92.51	91.30	93.73	87.21	90.15	81.26	85.43	91.29	92.17	78.64	88.43	84.79	88.17

Cross-view consistency alone does not guarantee a reliable revision. EC-CVEC therefore applies semantic preservation checks to prevent the candidate from dropping key entities, drifting to another metaphorical domain, adding unsupported details, or relying on weak reasoning evidence. Let C_{cv} denote the alternatives that satisfy both the cross-view and semantic constraints. The direct calibration candidate is selected as

$$A_c = \underset{a \in C_{\text{cv}}}{\operatorname{argmax}} W(a). \tag{6}$$

If C_{cv} is non-empty, A_c is used to revise the baseline prediction. Otherwise, EC-CVEC retains A_b temporarily and proceeds to option-permutation verification.

Option-Permutation Verification. For samples with unresolved disagreement, EC-CVEC considers the supported non-baseline candidates $C = \{a \in \mathcal{Y} \setminus \{A_b\} \mid W(a) > 0\}$. When $C \neq \emptyset$, EC-CVEC selects the highest-supported alternative $A_r = \operatorname{argmax}_{a \in C} W(a)$. Ties are resolved using the deterministic rule adopted in MS-MVR. It then applies forward and backward cyclic permutations to the candidate options and maps each prediction back to its original candidate identity.

Let $\Gamma_{\text{perm}} \in \{0, 1\}$ indicate whether a valid residual candidate exists and passes the bidirectional permutation verification and semantic preservation checks. Specifically, $\Gamma_{\text{perm}} = 1$ only when $C \neq \emptyset$, A_r receives sufficient votes, outperforms A_b in both permutation directions, and preserves the essential semantic content. The full multi-view reasoning process is not repeated during this stage, limiting the additional inference cost.

Final Decision. The final prediction is determined conservatively as

$$\hat{A}_{\text{final}} = \begin{cases} A_c, & C_{\text{cv}} \neq \emptyset, \\ A_r, & C_{\text{cv}} = \emptyset \wedge \Gamma_{\text{perm}} = 1, \\ A_b, & \text{otherwise.} \end{cases} \tag{7}$$

5 Experiments

Evaluation Protocol. We evaluate MERCURY against 18 representative open- and closed-source MLLMs on BiMeRe. Seven prompting settings are considered: *Zero-shot*, *CoT*, *Domain*, *Emotion*, *One-shot*, *Two-shot*, and *Three-shot*. Accuracy is used as the primary evaluation metric.

Training and Inference Details. BiMeRe is independently divided into training, validation, and test sets for Chinese and English using a 6:2:2 ratio. During Stage 1, MERCURY is trained on HAIVMet (3,267 samples) and MemeCap (5,823 samples). The visual encoder and language model are frozen, while the projector is optimized for 2 epochs with a learning rate of 1×10^{-3}. During Stage 2, the visual encoder remains frozen, while the projector and Vicuna are jointly optimized on BiMeRe for 5 epochs with a learning rate of 2×10^{-6}. We use a cosine learning-rate scheduler with a warmup ratio of 0.03.

Training is conducted on four NVIDIA A100 GPUs. MERCURY and the open-source baselines are evaluated on NVIDIA A40 GPUs, while closed-source models are accessed through their official APIs.

Baselines. We compare MERCURY with 18 representative MLLMs. The open-source baselines include VisualGLM-6B, Qwen-VL-Chat [2], InternLM-XComposer2-VL-7B [8],InstructBLIP-T5-XL [7], BLIP-2-Flan-T5-XL [16], Yi-VL-6B-Chat, Yi-VL-34B-Chat [29], MiniCPM-Llama3-V-2.5 [28], Idefics2-8B [15], DeepSeek-VL2 [23], InternVL2-8B [6], InternVL-2.5-8B, and Qwen2.5-VL-72B [3].

The closed-source baselines include GPT-4o [18], Qwen-VL-Max, Gemini-2.5-Flash [10], GLM-4V-Plus [12], and Claude-3.7-Sonnet [1]. Together, these baselines cover different model scales and multimodal architectures, providing a broad comparison for bilingual multimodal metaphor understanding.

5.1 Performance Analysis

As shown in Table 1, MERCURY achieves the best overall accuracy on both language subsets, reaching 70.87% on English and 72.07% on Chinese. It outperforms Claude-3.7-Sonnet, the strongest English baseline, by 4.67 percentage points, and Qwen-VL-Max, the

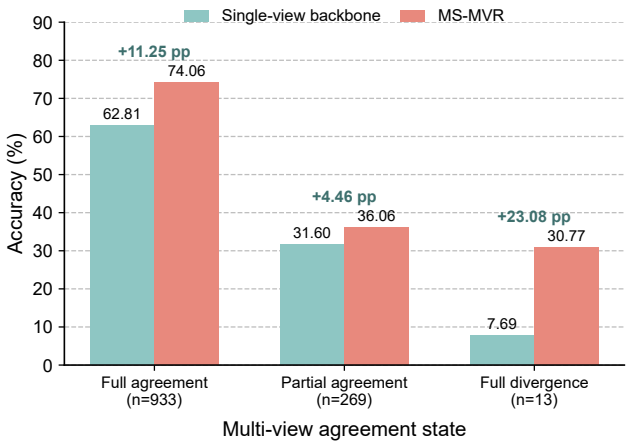


Figure 4: Performance comparison between the single-view backbone model and MS-MVR under different multi-view consistency states.

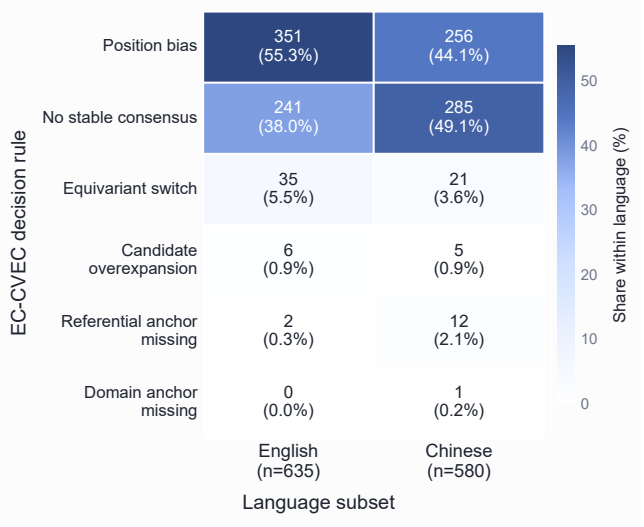


Figure 5: Rule trigger distribution of EC-CVEC on the Chinese and English test subsets.

Table 2: Ablation results of key modules in MERCURY.

Setting	EN ACC(%)	ZH ACC(%)
MERCURY (Ours)	**70.87**	**72.07**
w/o EC-CVEC	64.57	65.86
w/o MS-MVR	60.31	57.24
w/o Both (Vanilla Fine-tuning)	55.59	55.00

strongest Chinese baseline, by 5.15 percentage points. For a fairer comparison, we further align the evaluated sample IDs across models: Claude-3.7-Sonnet achieves 67.30% on English, while InternVL-2.5-8B achieves 69.66% on Chinese. MERCURY still exceeds them by 3.57 and 2.41 percentage points, respectively, indicating that its gains are not caused by differences in test-sample composition.

5.2 Ablation Study

Table 2 evaluates the contribution of the two main components in MERCURY. Removing EC-CVEC reduces accuracy from 70.87% to 64.57% on English and from 72.07% to 65.86% on Chinese, corresponding to drops of 6.30 and 6.21 percentage points, respectively. Removing MS-MVR causes larger decreases of 10.56 points on English and 14.83 points on Chinese, demonstrating the importance of complementary multi-view evidence for metaphor interpretation. When both modules are removed, the vanilla fine-tuned model achieves only 55.59% and 55.00%, confirming that MS-MVR and EC-CVEC provide complementary improvements beyond task-specific fine-tuning.

5.3 Mechanism Analysis under Multi-View Consistency

To examine how MS-MVR benefits different evidence-consistency regimes, we group test samples into full consistency, partial consistency, and complete disagreement according to the predictions of different views. As shown in Figure 4, MS-MVR consistently outperforms the single-view backbone across all three groups. For the 933 fully consistent samples, accuracy increases from 62.81% to 74.06%, a gain of 11.25 percentage points, indicating that MS-MVR effectively reinforces mutually supportive evidence. For the 269 partially consistent samples, accuracy improves from 31.60% to 36.06%, showing that it can still aggregate useful evidence under local conflicts. The largest improvement is observed on the 13 completely disagreeing samples, where accuracy rises from 7.69% to 30.77%; however, this result should be interpreted cautiously because of the small sample size. Overall, these findings suggest that MS-MVR improves prediction reliability both when reasoning views agree and when their evidence is partially or fully conflicting.

5.4 Analysis of the EC-CVEC Rule-Triggering Mechanism

Figure 5 shows that EC-CVEC primarily retains the baseline prediction because of display-position bias or the absence of stable equivariant consensus. These rules cover 55.3% and 38.0% of the English samples, respectively, compared with 44.1% and 49.1% of the Chinese samples, suggesting different uncertainty patterns across the two languages. EC-CVEC switches only 35 English and 21 Chinese predictions, accounting for 4.61% of all 1,215 test samples, which confirms its conservative behavior. Among these 56 switched samples, 46 produce correct final predictions, corresponding to a correction success rate of 82.14%. Semantic constraints, including expansion, referential-anchor, and domain-anchor checks, trigger less frequently but prevent weakly supported alternatives from replacing the baseline. Overall, EC-CVEC improves prediction reliability through limited, semantically grounded, and permutation-stable revisions rather than large-scale answer rewriting.

6 Conclusion

This paper addresses the limitations of existing multimodal metaphor datasets, which are largely monolingual and focused on coarse-grained recognition, by constructing BiMeRe, a Chinese-English bilingual benchmark for fine-grained metaphor interpretation. To mitigate reasoning instability caused by local visual cues, misleading distractors, and option-position bias, we further propose MERCURY, which acquires complementary evidence through metaphor-specialized multi-view reasoning and performs conservative prediction correction through equivariance-constrained calibration. Extensive experiments against representative open- and closed-source MLLMs, together with ablation and mechanism analyses, demonstrate the effectiveness of structured evidence aggregation and stability-constrained calibration for abstract and non-literal multimodal understanding.

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